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Recent advances in SiC via etch process for RF device manufacture

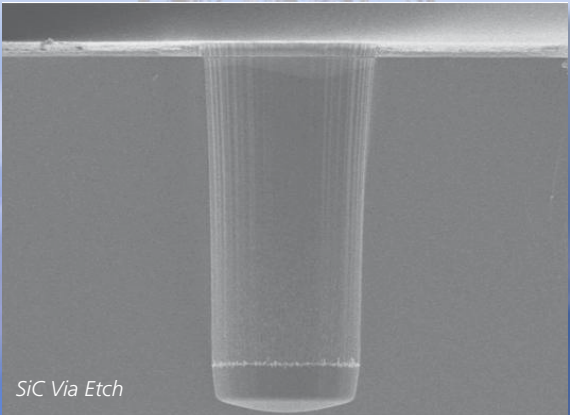
Plasma process solutions for RF device manufacture

As a leading provider of process solutions for a broad range of applications, we're delighted to announce the development and launch of the SiC via plasma etch process using our high performance **PlasmaPro®100** Polaris® etch system.

SiC is becoming an increasingly important material, particularly in the arena of high performance GaN RF devices using SiC as a substrate. A smooth via etch through the SiC is essential to enable these devices, and Oxford Instruments has developed the ideal solution for etching high quality SiC vias efficiently. Combined with a low damage GaN etch within the same hardware, the **PlasmaPro 100** Polaris offers a unique capability for GaN based RF device plasma etch processing requirements.

The technology developed offers several process capabilities suited to the SiC via application:

- High SiC etch rate enabling maximum throughput
- Smooth sidewalls for problem free post etch metallisation
- High selectivity to underlying GaN layer giving a smooth, low damage stop onto the GaN device layers



- Clamping of sapphire carriers using Oxford Instruments' unique patented Electrostatic Clamp technology ensuring excellent sample temperature control and maximum yield
- Capability of etching SiC and GaN in the same tool through advanced plasma source technology
- High utilisation provided by long Mean Time Between Cleans (MTBC)

Dr Mark Dineen, Optoelectronics Product Manager at Oxford Instruments Plasma Technology says, "Our Applications specialists have spent significant time developing this SiC via etch process on the **PlasmaPro 100** Polaris etch system, enabling high selectivity and throughput amongst other benefits. These benefits will enable our customers to etch both SiC and GaN in the same tool through advanced plasma source technology."



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High quality MoS₂ CVD growth process developed

Growth of Molybdenum Disulphide (MoS₂) & related 2D TMDs

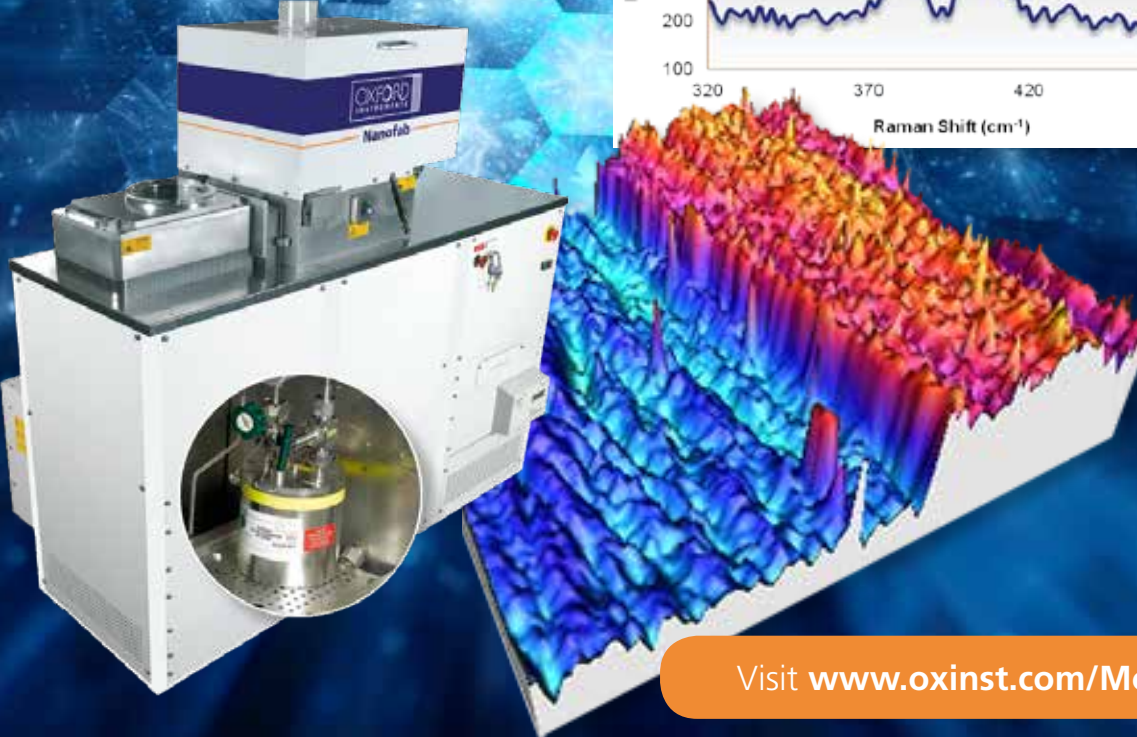
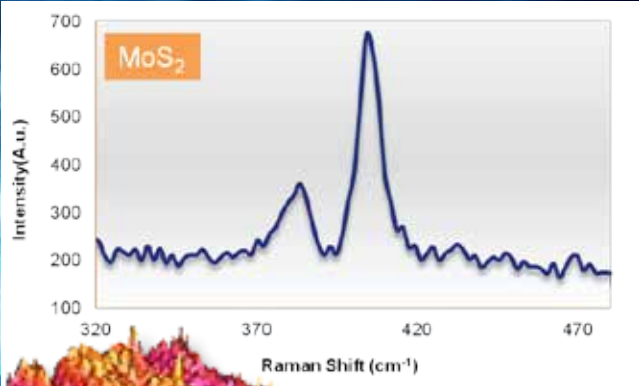
We recently announced the development and launch of the MoS₂ growth process using our high performance **Nanofab®** nanoscale growth system.

Single layer MoS₂ is a direct band gap semiconductor which has wide ranging applications in optoelectronics such as LEDs, photovoltaics, photodetectors, and bio sensors while multi layer MoS₂ is an indirect band gap semiconductor which shows promise in future digital electronics.

Our scientists have undertaken extensive research and optimisation of this CVD process, developed on a Nanofab system equipped with precursor delivery modules capable of delivering a wide range of liquid/solid/metal organic precursors suitable for 2D materials growth. Offering growth on a range of substrates including sapphire and ALD alumina (Al₂O₃),

the system is capable of depositing other 2D transition metal dichalcogenides (TMDs) such as WS₂, MoSe₂ etc.

This process development and its proven results are extremely exciting, as we enter a new phase in our 2D materials processing capabilities with the **Nanofab** plasma processing system. Raman analysis has demonstrated a high quality mono-layer, with AFM and Raman maps showing resultant smooth and uniform films. We anticipate that this development in 2D materials growth will facilitate the next generation of nano electronic devices.



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In-situ plasma pre-treatments on GaN surfaces for reduction of interface traps in metal-oxide-semiconductor capacitors & high electron mobility transistors using ALD Al_2O_3

J.W. Roberts, University of Liverpool

P.R. Chalker, S.J. Cho, I.G. Thayne, K.B. Lee, Z. Zaidi, P.A. Houston

GaN is regarded as a promising semiconductor for next generation power electronic devices due to its high critical breakdown field, high maximum operating temperature, and high switching speeds^{1,2}.

Gate dielectrics used in insulated gate transistors based on GaN also need to show good breakdown properties. Atomic layer deposited Al_2O_3 with its wide band gap, low leakage and high breakdown field is an excellent candidate for this purpose³. However, when Al_2O_3 is directly deposited onto GaN and AlGaIn surfaces, it is associated with a high interface trap density of the order of $10^{13} \text{ cm}^{-2}\text{eV}^{-1}$ and higher⁴. These interface states give rise to large hysteresis, threshold voltage shifts and trap charging effects which adversely affect device operation⁵.

Here we report the effect of in-situ surface treatments using Ar and N_2 remote plasmas on the GaN surface prior to dielectric deposition. All surface treatments and Al_2O_3 depositions were carried out on an Oxford Instruments **OpAL** Plasma ALD system. GaN metal-oxide-semiconductor (MOS) capacitor samples were cleaned with acetone and isopropyl alcohol before introduction into the chamber at 200°C.

Three sets of devices were processed:

- A control sample with no plasma surface treatment
- N_2 plasma treated samples
- Ar plasma treated samples

The plasma treatments consisted of 5 minutes at 50W, 150W or 300W using a gas flow rate of 60sccm. For each device 200 cycles of Al_2O_3 were deposited from trimethyl aluminium (TMA) and H_2O at 200°C using the following process: 0.02s TMA / 3s Ar purge / 0.02s H_2O / 3s Ar purge. GaN metal-insulator-semiconductor high electron mobility transistors (MISHEMTs, Figure 1) were also processed using the best results for the Ar and N_2 plasma treated samples.

Figure 2 shows the results of GaN MOS capacitor stress testing. The dashed black line on each graph shows the results for the control sample. With the application of either N_2 or Ar plasma surface treatments flatband voltage (V_{FB}), hysteresis, frequency dispersion and interface state density (D_{it}) are all reduced for plasma powers ranging from 50W to 300W. For the Ar plasma treatment the best results were found for the 50W process, including a reduction in D_{it} by 3 orders of magnitude. Using the N_2 plasma treatment showed similar improvements with the best results found for the 150W process. Comparison between the Ar and N_2 plasma surface treatments shows that the 150W N_2 process gave the lowest flatband voltage shift and hysteresis while the 50W Ar process gave lower D_{it} and frequency dispersion.

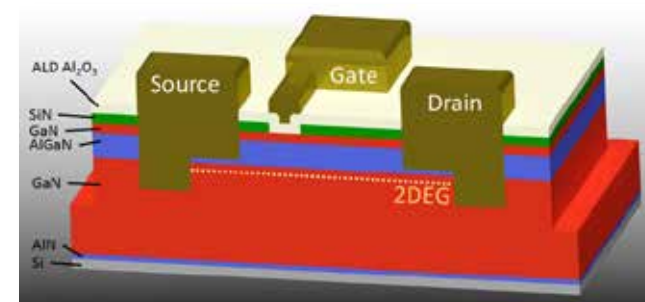


Figure 1: X-section diagram of GaN power transistor

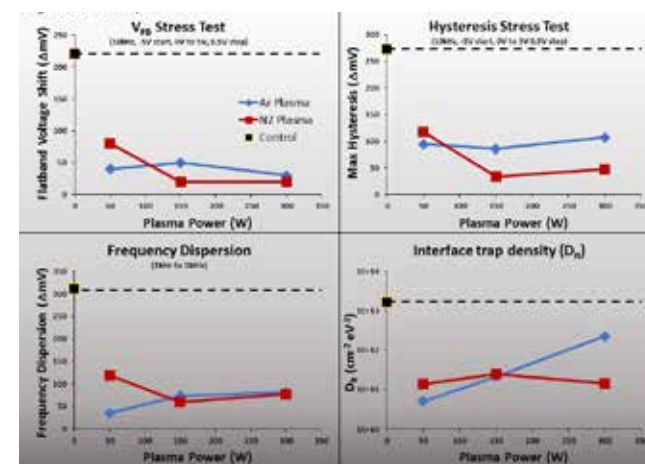


Figure 2: MOS capacitor results

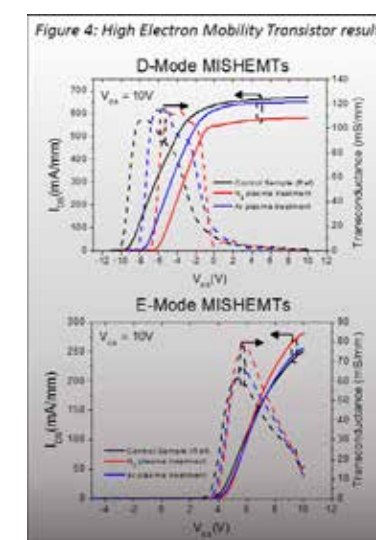


Figure 3: High electron mobility transistor results

Figure 3 shows results of the 50W Ar and 150W N_2 plasma surface treatments on normally off (D-mode) and normally on (E-mode) MISHEMTs. In the D-mode device the threshold voltage (V_{TH}) was shifted more positive and the peak transconductance was improved. The source-drain current in the on-state was reduced compared to the control sample. In the E-mode device a similar increase in transconductance was observed along with an increase in the source-drain current. We postulate that these improvements in performance in both the MOS capacitors and the MISHEMT devices can be attributed to either the removal of native GaO_x before the deposition of the gate oxide, reducing the number of interfaces near the GaN surface or from the formation of higher quality GaO_x with fewer electronic traps. Research is on-going to determine the mechanisms behind these improvements.

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Acknowledgements:

This work has been supported by the Engineering and Physical Sciences Research Council Programme Grant EP/K014471/1 "Silicon Compatible GaN Power Electronics".

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Precision 3-D nanomachining of silicon nanowires

Katarzyna Korwin-Mikke^a, Mark McNie^a, Erwin Berenschot^b, Roald Tiggelaar^c, Meint De Boer^c
This article is based on a paper presented at MNE 2016 in Vienna (Sept. 2016)

Silicon is the most common material used in semiconductor fabrication. Due to its properties, availability and ease of processing, it is widely used in areas such as electronics, MEMS, NEMS and photonics.

Increasing attention is focused on nanoscale etching for silicon including the formation of advanced geometry transistor gate (e.g. FinFET), trench isolation, photonic crystals and nanowires. Nanowires have potential applications in photonics^[1], photovoltaics^[2], optoelectronics^[3] and microelectronics^[4]. The focus of this work was to demonstrate the formation and control the shape of arrays of nanowires at the wafer scale using a top down fabrication approach – employing a combination of Displacement Talbot Lithography (DTL)^[5] and plasma etch.

A mixed gas nanoetch process^[6] based on a SF_6 - C_4F_8 chemistry (also known as the pseudo-Bosch process) was selected for this work. It continuously passivates the sidewalls for anisotropy whilst the etch front is kept clear by accelerated ions at a room temperature - offering controlled etch rate and good sidewall quality (without the scallops that are a characteristic of the Bosch process). The etch profile is also independent of the crystalline orientation of the silicon. It is somewhat limited in selectivity and aspect ratio compared to the cryogenic process, particularly as etch openings shrink.

100mm silicon wafers were patterned with a 30 x 30mm² array of nanodots (100nm diameter and 200nm thick PR/BARC (PFI88/AZBarli-II200) using DTL to realise a mask for nanowire fabrication. This non-contact proximity exposure method uses a phase-shift line pattern mask and two orthogonal exposure cycles, followed by development and 1:1 transfer of the PR pattern into the BARC with N_2 -based directional plasma etching to realize the nanodots. It is lower cost and more rapid than direct write alternatives for patterning at the nanoscale.

A dry etch was developed on our **PlasmaPro 100** Estrelas system to transfer the nanodots into an array of 100nm wide nanowires on a 250nm pitch. The mask was slightly conical and so the sloped profile translated down into the final structure as the mask eroded and the edge of the mask pulled back during an anisotropic etch, narrowing the top of the nanowire slightly. The higher the selectivity, the less the mask pulls back and also the deeper an etch can be performed for a given mask thickness. The baseline process developed had a selectivity >5:1 to the resist (Figure 1) to etch nanowires to a depth of 440nm in single stage process and a vertical etch rate of 125nm/min with a depth uniformity $\leq \pm 1\%$. The resulting profile was almost vertical ($\sim 89^\circ$). This could be improved by optimising the mask initial profile or by etching using a hard mask (e.g. oxide or alumina) with higher selectivity. A thicker mask or higher selectivity could also potentially yield deeper nanowires.

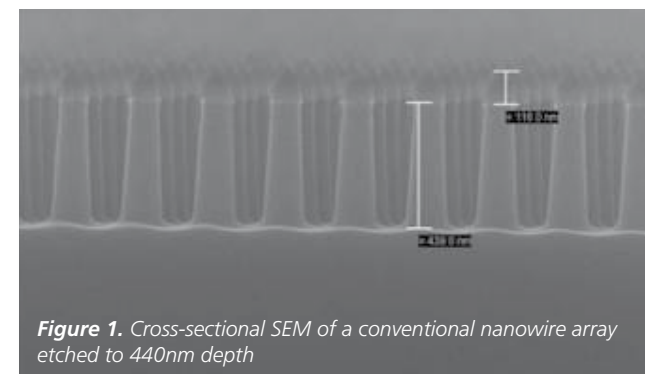


Figure 1. Cross-sectional SEM of a conventional nanowire array etched to 440nm depth

To realise shaping in the third dimension, the width of the nanowires was modulated by tuning etch-passivation gas ratio against the bias power (controlling the energy of the accelerated ions at the etch front) as the etch proceeded. Demonstration of single and double width modulations was targeted (Figure 2) to narrow nominally 100nm diameter nanowires to around 50nm. The single modulation was developed by adding an under-passivating stage to decrease the nanowire width, a balanced process to realize a thinned vertical section and then an over-passivating stage to restore the nanowire width. The

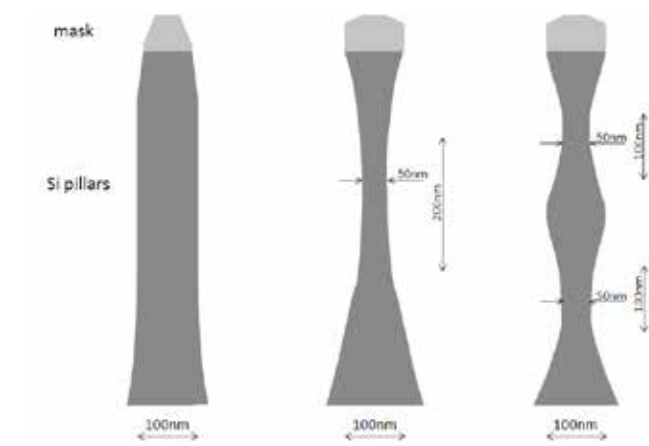


Figure 2. Sketch of target example structures in nominally 100nm diameter nanowires

degree of thinning, its location and length of thinned region may all be controlled by adjusting the process. Figure 3 shows a single modulation with a 53nm thinned region of length 226nm. The overall nanowires were etched to 484nm with an etch rate of 186nm/min and selectivity to the mask >5:1. To achieve double modulation, additional steps were taken to thin and increase the width of the nanowire twice. The modulation length was reduced to achieve this within a similar depth, 460nm, (Figure 4) with a vertical etch rate 118nm/min and a selectivity of 9:1 and uniformity $\leq \pm 1\%$.

Future work will seek to increase the depth by optimizing the mask and investigate tuning of transitions achievable in the process as well as generation of more complex three-dimensional nanowires.

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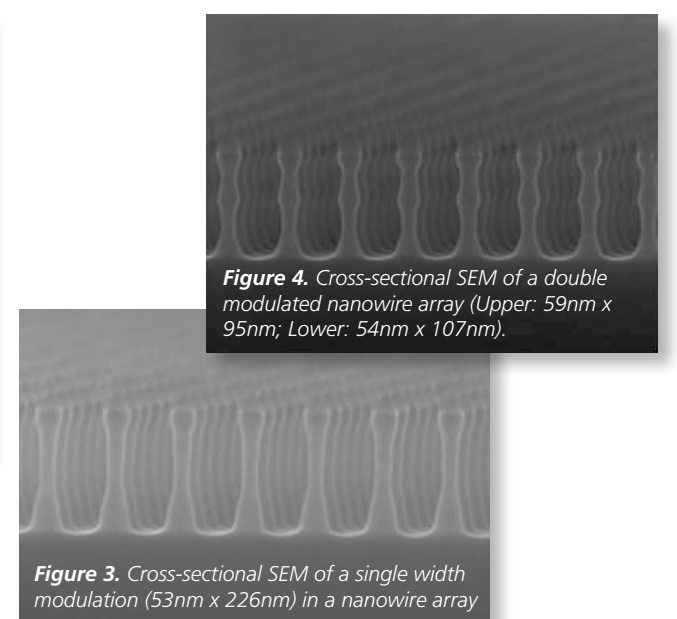


Figure 4. Cross-sectional SEM of a double modulated nanowire array (Upper: 59nm x 95nm; Lower: 54nm x 107nm).

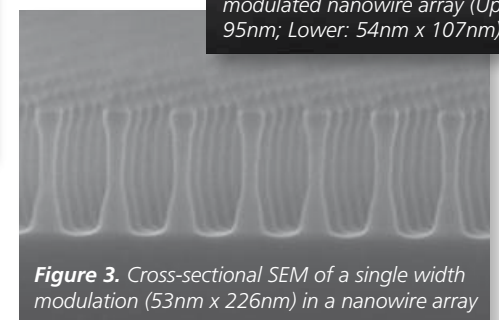


Figure 3. Cross-sectional SEM of a single width modulation (53nm x 226nm) in a nanowire array

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One dimensional contacts to a 2D material

Dr Gregory Auton, School of Computer Science, The University of Manchester

Graphene is a 2D material with a unique set of properties that make it extremely desirable in industry for a wide range of applications^[1].

Despite the relatively short period of time since its discovery some applications have already seen some success. For example, its extreme mechanical and thermal properties make it perfect for composite materials and its transparency and high conductivity have led to its use in touch screen technology. However, certain limitations have blocked its use in some industries, most notably in electronics where the lack of band gap and high contact resistances have meant that there has been little progress. Despite this, a great deal of research has focused on these problems because the rewards of using graphene are significant. As the material with the highest room temperature carrier mobility, it could be used to fabricate considerably faster field effect transistors (FETs) than traditional materials.

Multiple potential solutions have been found to the problem of having no band gap such as: using graphene nano ribbons, introducing a tunnelling barrier or by modifying the graphene chemically. However, until recently very little progress had been made trying to improve the contact resistances of graphene devices. Indicial experiments proved that conduction through the Basel plane was minimal^[2] and further experiments showed the importance of the edge of graphene^[3]. It was not until the introduction of stamp transfer that it became possible and necessary to contact the edge and only the edge of the graphene^[4]. Using boron nitride (BN is an insulating 2D crystal) as a stamp, graphene could be encapsulated using the Van Der Waals interactions. Since the graphene did not touch any hydrocarbon polymers or solvents, the highest quality graphene to date was fabricated with atomically clean interfaces. However, there was no way to contact the graphene since both

Basel planes were covered in insulators, so an edge had to be formed to make contact to. Using a CHF₃/O₂ plasma it was possible to etch both the BN and the graphene simultaneously

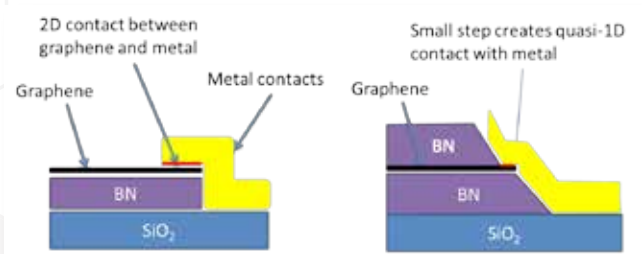


Figure 1. a schematic demonstrating both 2D contacts (left) and 1D contacts (right).

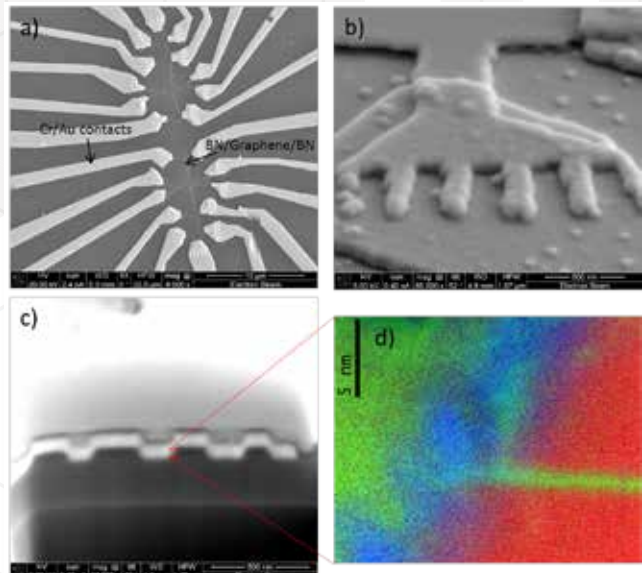


Figure 2. a & b: SEMs of an early device with 1D contacts. c: Ion milling has been performed to create a crosssectional

exposing a 1D line of graphene to make contact (demonstrated in fig. 1 and 2). It was demonstrated that these 1D contact were of higher quality than anything that had gone before them^[4]. However, in this work^[4] the etch masks could not be used for lift off once an etch had been performed meaning that a second lithography step was necessary to do metalisation. This in turn meant that the 1D contact was contaminated with

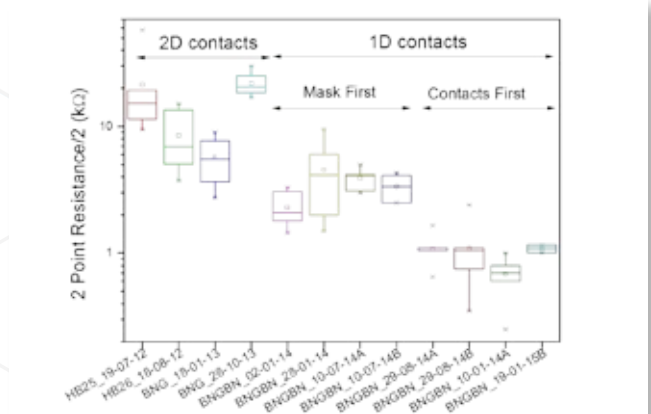


Figure 3 a graph demonstrating the 2 point contact resistances divided by 2 for various graphene devices made using different techniques.

hydrocarbons, the very latest techniques developed allow the same mask to be used for the etch as the lift off. Effectively allowing the contacts to be placed before the geometry of the device is defined with a separate mask. This contact first technique gives a perfectly self-aligned atomically clean metal-graphene interface for low resistance ohmic contacts (figure 3 demonstrates the improvement of contact resistances). It is also important these are self aligned to reduce capacitive coupling between the graphene and any overlapping metal through the BN. Any capacitive contributions from the contacts are highly detrimental to any high frequency application and must be minimised.

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Plasma ALD of SiO₂, NiO and HfO₂ on the FlexAL: Modifying Flow, Pressure and Plasma Parameters

Agnes Kurek¹, Saleem Shabbir¹, Tom Miller¹, Aileen O'Mahony¹, Harm Knoops^{1,2}, Annika Peter¹, Owain Thomas¹, & Bob Gunn¹

Atomic layer deposition (ALD) of ultra-thin oxide films is critical for maximum device performance in many applications including photovoltaics¹, optics², and microelectronics³. Within a robust plasma ALD process, the elements of flow, pressure and plasma conditions must be optimised to achieve best film quality. The effects of these parameters on the RI, VBD and Uniformity of newly-developed ALD processes were presented recently at by Dr Agnieszka Kurek our ALD Process Engineer at the 2016 BALD Conference.

Silicon dioxide films have been demonstrated by plasma ALD using bis(tert-butylimino)silane (BTBAS) and oxygen plasma. The combination of a high working pressure controlled by a quick-action APC valve with the rapid pumping of a turbomolecular pump, has increased the film deposition rate (nm/min), and growth per cycle, without compromising film quality in terms of electrical and optical properties. This process is robust, repeatable and conformal on high aspect ratio structures, up to 30:1.

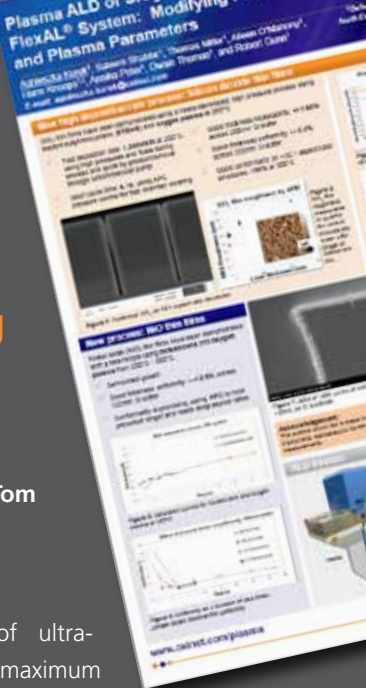
Nickel oxide films have been demonstrated by plasma ALD, using nickelocene and oxygen plasma. Rapid bubbling of the precursor delivers efficient dosing, while low pressure O₂ plasma allows for fast surface saturation and good uniformity.

Hafnium oxide films have been demonstrated by plasma ALD using tetrakis(dimethylamido)hafnium (TDMAH) and oxygen plasma. This results in improved deposition uniformity over tetrakis(ethylmethylamido) hafnium (TEMAH), even at short cycle times. Electrical characterisation of resultant HfO₂ films will be discussed.

The important parameters for plasma ALD of SiO₂, NiO and HfO₂ were discussed, focusing on the effect of flow and pressure for each process step.

For the full poster please contact: plasma@oxinst.com

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PROCESS

Resistance repeatability study of ion-beam deposited vanadium oxide thin films

Pauline Alvarez, Applications Engineer, Oxford Instruments Plasma Technology
This article is based on a paper presented at SPIE 2016 in San Diego, USA (Aug. 2016)

Ion Beam Sputter Deposition (IBSD) is a versatile technique particularly suited to applications requiring high quality, high performance layer materials as it allows independent and accurate control of the process parameters.

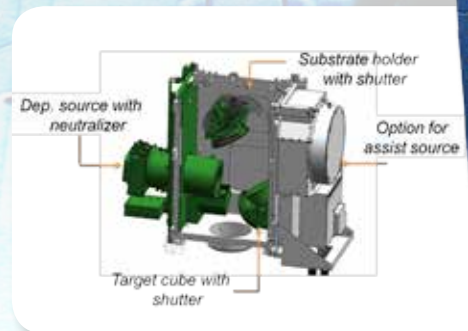
Vanadium oxides, used for example in the fabrication of microbolometers, optical switches or optical storage, exhibit interesting properties such as a high Temperature Coefficient of Resistance (TCR), relatively low 1/f noise and a semiconductor-metal phase transition close to room temperature. However, it is very challenging to control the stoichiometry of the deposited film as there are at least 25 different oxidation states of vanadium, few of which display the required electrical characteristics.

In a recent study, vanadium oxide thin layers were deposited by IBSD using an Oxford Instruments **Ionfab300** ion beam system

and analysed with regard to their electrical properties.

The impact of the system parameters on the resistance repeatability, wafer-to-wafer and batch-to-batch, was thoroughly investigated to provide the end user with a clear understanding of the factors affecting film resistivity while ensuring at the same time a steep variation of resistance with temperature, as notably required for uncooled bolometers. These parameters were balanced to also achieve a good deposition rate, throughput and uniformity over large device areas, compatible with the requirements of industrial applications.

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Ionfab300 with brooks MMX handler

Prestigious Nanjing University orders several plasma systems

We recently won an order from Nanjing University of Posts and Telecommunications, based in Nanjiang, Jiangsu, China for multiple plasma etch systems to be used for silicon and III-V etching.

Our highly configurable **PlasmaPro 100** systems offer an extensive range of processes, making them eminently suitable for the nanotechnology research being undertaken by Nanjing University. **PlasmaPro 100** process modules are built on 200mm platforms, with single wafer and multi-wafer batch capability. The process modules offer excellent uniformity and high throughput processes on a range of applications.

"As a leading research centre in China we conducted a rigorous tender process and decided on Oxford Instruments for their state-of-the-art processing equipment that is key to our successful research", said Dr. Huang Xiaoming of the Nanjing University of Posts and Telecommunications, "The process modules offer excellent uniformity and high throughput processes, which together with a global customer support network, and low cost of ownership made this the system of choice for our University."



We'd like to introduce our new MD, Richard Pollard



Richard Pollard has recently joined our company as Managing Director

Most recently Richard was with Ocean Optics, Inc (Halma plc), a market leader in spectrometers and light sources, where he was President based in Florida, USA. Having previously held several senior leadership roles at Halma plc, in the USA and UK, Richard has a proven track record of delivering results in technology businesses, making him particularly well placed to lead Oxford Instruments Plasma Technology.

"Richard brings his already extensive experience in the high technology capital equipment sector to lead and develop Oxford Instruments Plasma Technology", comments Ian Barkshire, CEO of the Oxford Instruments Group, "As our markets expand, Richard will ensure the continued investment in new technology and applications in order that our customers remain at the forefront of their field. We are delighted that he has joined our team."

Richard is a Chartered Engineer, with a BEng Manufacturing Engineering from Brunel University and an MBA from the Open University Business School, Cambridge.

Optical emission monitoring for optimisation of atomic layer etch (ALE) processes

Andy Goodyear & Dr Mike Cooke, Oxford Instruments Plasma Technology
This article is based on a paper presented at MNE 2016 in Vienna (Sept. 2016)

ALE is becoming increasingly popular for the etching of thin layers with low damage, high precision and excellent selectivity. It has particular relevance to the etching of 2D materials.

We have developed an optimised etch tool for well controlled ALE. This includes fast gas dosing for accurate delivery of dosing gas, combined with precise delivery of RF power to the substrate for optimum control of DC bias^[1].

Switching between ALE and conventional etching is controlled from software, allowing ALE and conventional etch modes in the same process recipe.

We report the use of optical emission spectroscopy (OES) to highlight the potential problems of using a conventional plasma etch tool for ALE, and to demonstrate accurate dosing using the new hardware, as shown in Figs. 1-2. The gas decay time is also found to be shorter when using the ALE hardware, approximately 4secs, while for a conventional etch tool using a standard MFC it is over 15secs, as shown in Fig 3.

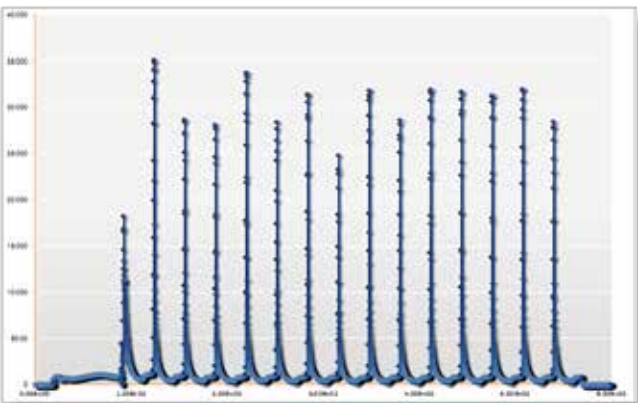


Figure 1. Variability in Cl OES intensity observed when using 2sec gas pulses in a conventional etch tool, due to MFC turn on time scatter and control system timing delays.

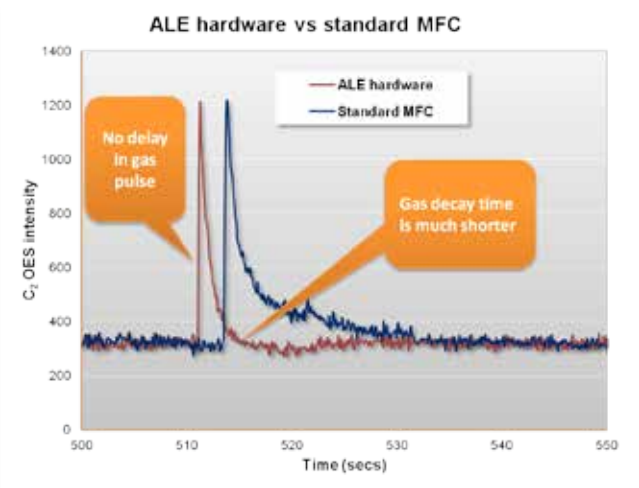


Figure 3. Improvements in gas pulse timing and gas decay time when using optimised ALE hardware.

OES was found to be an excellent tool for optimisation of the ALE process sequence, as it enables the monitoring of dose pulses, purging efficiency and etch species. OES can therefore be useful in minimising the number of process runs needed for ALE process development. Contaminant gases can also be monitored, enabling the user to identify potential sources of contamination and to optimise chamber preparation.

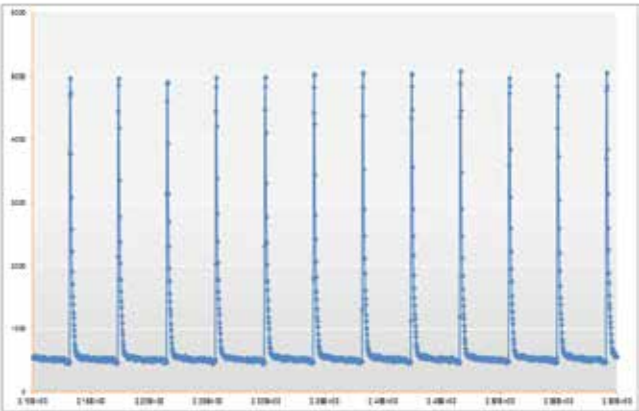


Figure 2. Improved stability with optimised ALE hardware.

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ICP CVD advanced plasma cleaning

Dr Owain Thomas, Applications Team Leader, Oxford Instruments Plasma Technology

Getting the maximum tool utilisation during deposition of dielectric films requires minimising the clean overhead. Here we report the details for SiO₂ but it is also equally effective for Si₃N₄ films.

The in-situ chamber plasma cleaning process plays an important role in the manufacturing throughput. In ICPCVD processing, a significant proportion of the tool time is devoted to running the plasma using etching gases to clean the chamber. This includes cleaning the inner ICP tube together with the chamber walls and the lower electrode. Previously the plasma clean process would be carried out after a significant amount of material was deposited in the chamber. A reduction in overall plasma clean time was therefore required in order to improve wafer throughput and cost of ownership of the system.

We have developed a new plasma clean process which has increased the system utilisation to >70%. The new plasma clean uses SF₆ and N₂O hence avoiding any O₂ which allows rapid change from deposition to cleaning without the fear of SiH₄ reacting with residual oxygen. Optical emission spectroscopy was used to monitor the F species during the plasma clean. Fig. 1 shows the difference in end point traces between our new and previous standard plasma clean processes. The new plasma clean results in 68% reduction in plasma clean time compared to the standard clean. The new Plasma clean also results in increased optical intensity with greater endpoint signal/resolution.

Further improvement in system utilisation was achieved by adopting a new interleave plasma clean mode. The new interleave plasma clean mode results in the plasma clean being carried out during the unloading of the sample. The actual plasma clean time depends on the film material, film quality and film thickness.

Fig. 2 shows a typical deposition/plasma clean sequence resulting in >70% utilisation based on a single 1micron film deposition followed by the interleave plasma clean. A thinner deposition (e.g. 200nm) results in a shorter clean step. This is based on depositing ICPCVD SiO₂ films with a typical specification as shown in table 1.

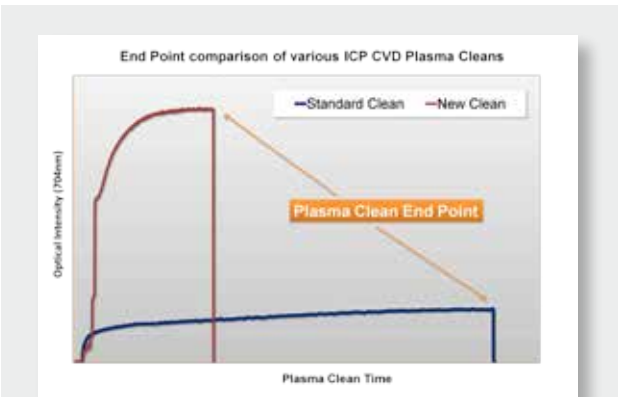


Figure 1: End point results of various ICPCVD Plasma Cleans

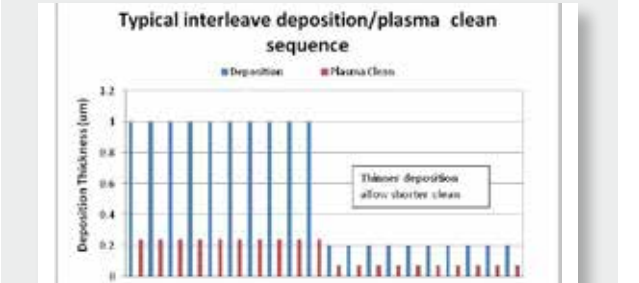


Figure 2: Typical interleave deposition/plasma clean sequence

Utilisation	> 70%
Deposition rate repeatability	< +/- 3%
Uniformity (over 150mm)	< +/- 3%
Film Stress	< - 100MPa
Particle Performance	<50 adders at >0.5um over 150mm wafers

Table 1: Typical ICPCVD SiO₂ specification

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PROCES

NEWS

Providing process and system solutions for the growing BioMEMS market

Frazer Anderson, *Strategic Marketing & Business Development Director*,
Oxford Instruments Plasma Technology

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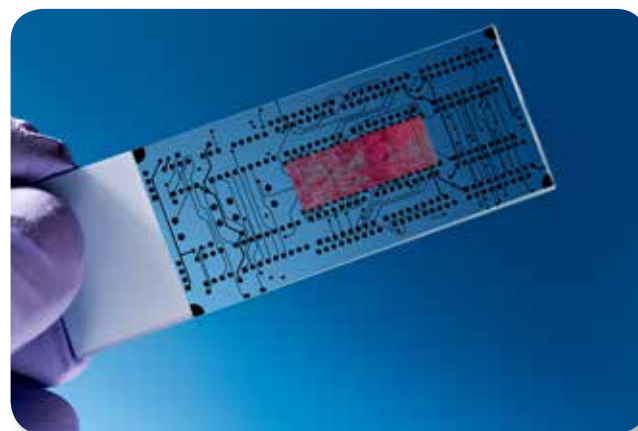
There are a number of obvious challenges! Moving into any new area requires us to gain an understanding of our customers' needs, which can, and often does involve learning a new "language". A language in which they are conversant and we are not! BioMEMS, and indeed any applications space

which involves biological or medical components offers a special challenge to those of us whose background is in the physical sciences.

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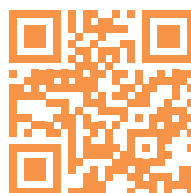
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